CS4680 Prompt Engineering
AI Assistant Project

Due Date
See the date in Canvas

Score
50

Questions and Directions

In this project, you will take on the role of an AI entrepreneur, using Generative APIs or open-source large language models (LLMs) to create a real-world AI-powered application. You are free to choose any platform for the Generative AI API, such as **OpenAI API, Gemini API, Meta API, or any other vendor**. Likewise, there are **no limitations** on which LLM technology to use. The goal is to apply your prompt engineering skills to solve a practical problem and build a minimum viable product (MVP) that demonstrates how AI can create value for users.

Project Requirements:

1. **Identify a Problem or Market Need:**
   Start by identifying a real-world problem or an unmet need where AI could provide a solution. Below are some examples of project ideas to inspire your work:
   - **Class Scheduler App**: Build an AI-powered app that optimizes course schedules based on preferences and commute times.
   - **AI Resume Reviewer**: Create an AI tool that reviews resumes and provides feedback based on job descriptions.
   - **Mental Health Chatbot**: Develop an AI chatbot that offers personalized mental health advice and resources.
   - **Study Buddy App**: Design an AI study assistant that generates personalized study plans and quizzes.
   - **AI Meal Planner**: Build an AI app that creates meal plans based on dietary preferences, budget, and cooking skills.
2. **Design Your Application:**
   As part of your project, **you will go through the process of designing** the core features of your application, but there is no need to submit anything specific for this step. Focus on:
   - **Defining** the functionality of your AI solution so that it's clear what your application will do.
   - **Describing** the interaction flow between users and the AI component to ensure smooth and intuitive user experience.

    - **Clarifying** the role that the Generative API or LLM will play in generating responses, assisting users, or solving tasks, so that your AI integration is purposeful.
3. **Build Your AI-Powered MVP (Minimum Viable Product):**
   Implement a working version of your AI application that:
    - Uses prompt engineering techniques to enhance the AI's output.
    - Effectively integrates the Generative API or open-source LLM to deliver practical, valuable solutions.
    - Demonstrates how your application solves the chosen problem or provides value to users.
4. **Prompt Engineering Techniques:**
   Incorporate at least two prompt engineering patterns (e.g., Persona, Few-Shot, Iterative Refinement) from the course. Explain how these patterns are used to optimize the AI's performance and improve the user experience within your application.
5. **User Interface Design (UI):**
   A **user interface (UI) is required** for this project. The UI can be in any format, such as a **mobile app, web application, desktop software, or a game interface**. It should be user-friendly and intuitive, allowing users to easily interact with the AI system. The UI must clearly showcase the AI-generated outputs and include necessary input fields for users. Think about how the design enhances the overall user experience and accessibility of your application.
6. **Project Presentation (with Live Demo):**
   During the project presentation, **the most important part is to run a live demo of your application**. Your presentation should demonstrate:
    - The problem your application solves.
    - A live demo showing how your application works in real-time.
    - How you used prompt engineering techniques to enhance the AI.
    - The impact and potential future applications of your AI-powered product.
      The live demo will be a key part of the evaluation, so ensure your application is functional and ready for presentation.

Submission Guidelines:

- **Code**: Submit your code repository link (GitHub URL).
- **Shot Video**: Submit a short demo video link (e.g., YouTube URL).

Grading Criteria:

- Problem Identification and Relevance (10%)
- Application Design and Implementation (30%)
- Use of Prompt Engineering Techniques (20%)
- UI/UX Design (10%)
- Presentation and Live Demo (30%)

Short Video Demonstration and Submission

Each student will record a short video (less than 2 mins) of their final project during the last week of the semester. This is an opportunity to showcase your application's functionality, design, and the problem it solves. The video should be TikTok/Youtube Short style short videos, and it should short, fun, and engaging. Please publish the video.

Submission Requirements: each student must submit their project through a Google Form (https://forms.gle/nMHp5owwYidszwt58). The submission must include:

- A link to the GitHub repository containing the complete project code and documentation.
- A short video
- Publishing (Bonus): While not required, publishing your application (making it publicly accessible online) is highly encouraged and will be rewarded as a bonus. This could involve deploying your web application to a hosting service, publishing your mobile app on app stores, etc.